



CS & IT ENGINEERING



Algorithms

Analysis of Algorithms

Lecture No.- 01



By- Aditya sir

Topics to be Covered



Topic

Schedule

Topic

Outcomes

Topic

Intro to Algorithms



About Aditya Jain sir

77.33/100



1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science (TA)
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.



9:30 - 11:30
Pm

Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

TTTe



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1. Analysis of Algorithms

1. Algorithm Concept and Lifecycle

2. Analysis of Algorithms → why, what, How

3. Methodology & Types of Analysis → How

☆☆ 4. Asymptotic Notations

5. Framework for Analysing Recursive Algorithms

6. Apriori analysis of Non-Recursive Algorithms

7. Analysing Loops

8. Space Complexity

9. Mathematical Background

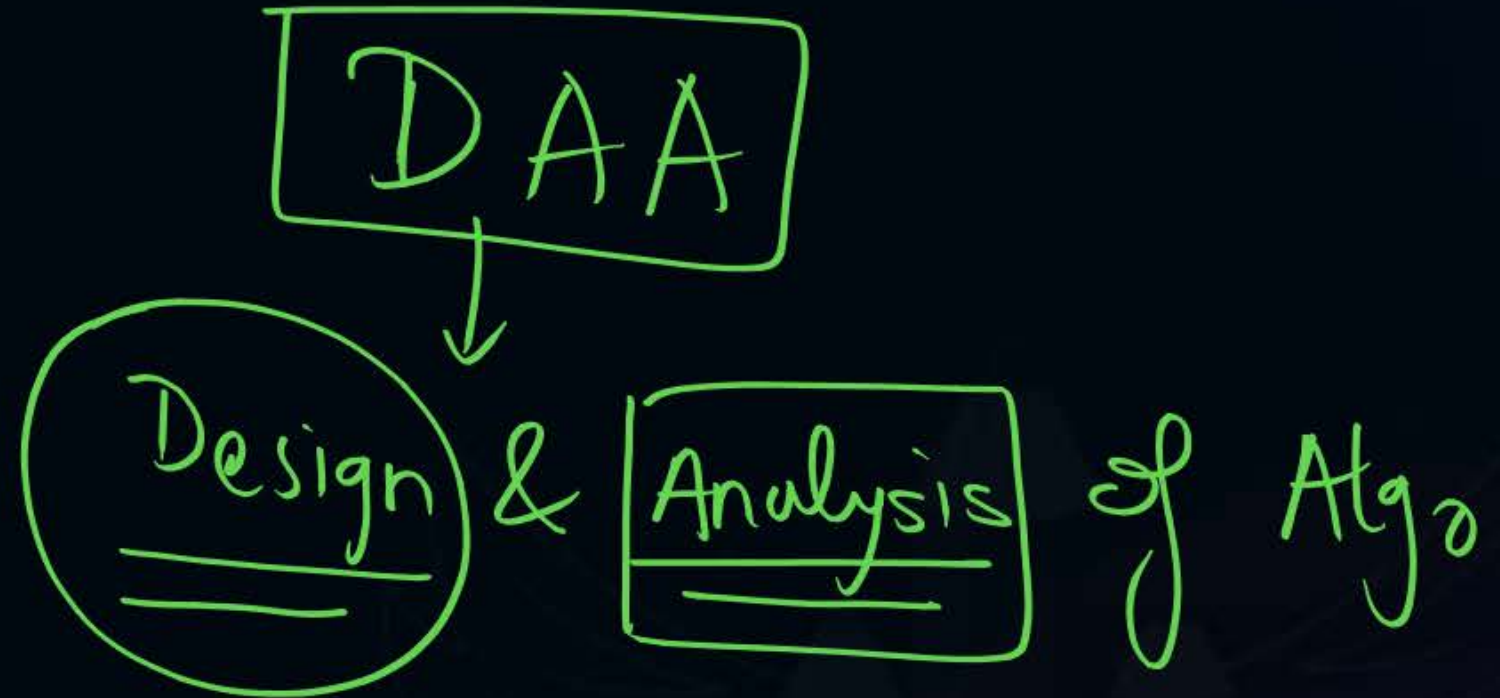


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8. Sorting Algorithms

1. Basic terminology
2. Methods
 1. Bubble Sort
 2. Selection Sort
 3. Insertion Sort
 4. Radix Sort





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2. Divide & Conquer (Design Strategies)

1. General Method
2. Max-Min Problem
3. Merge Sort
4. Binary Search
5. Quick Sort
6. Matrix Multiplication
7. Long Integer Multiplication (LIM)
8. Master Method for D and C Recurrences
9. Recursion Tree (end)

- ① Divide & Conquer
- ② Greedy
- ③ Dynamic Programming (DP)



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3. Greedy Method

1. General Method

2. Knapsack Problem (Fractional)

3. Job Sequencing with Deadlines (JSD)

4. Optimal Merge Patterns (OMP)

5. Huffman Coding

6. Minimum Cost Spanning Trees (MCST)

1. Prims Method

2. Kruskal's Method

6. Dijkstra's Shortest Paths Problem

(SSSP)

(V.V. Imp)



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4. Dynamic Programming (DP)

1. The Framework
2. Difference between DP, Greedy Method and Divide and Conquer

3. Multistage Graphs

4. Travelling Salesperson Problem (TSP) → Floyd Warshall

5. Binary Knapsack Problem / All Pairs Shortest Paths (APSP)

7. Bellman-Ford Single Source Shortest Paths (SSSP)

★ 8. Longest Common Subsequence (LCS)

★ 10. Matrix Chain Multiplication ^{(m(m))} / sum of Subsets (SOS)

11. Optimal Cost Binary Search Tree



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5. Graph Algorithms

1. Representation of Graphs
2. Graph Traversals

DFS

- 5.2.1 Undirected Connected Graphs
- 5.2.2 Undirected Disjoint Graphs: DFT
- 5.2.3 Directed Graphs & Types of Edges
- 5.2.4 DAG

BFS

- 5.2.5 FIFO BFS
- 5.2.6 LIFO BFS

Applications

of DFS & BFS



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6. Heap Algorithms (min Heap + Max Heap)
1. Operations : Create, Insert, Delete, Modify
 2. Applications : Heapsort



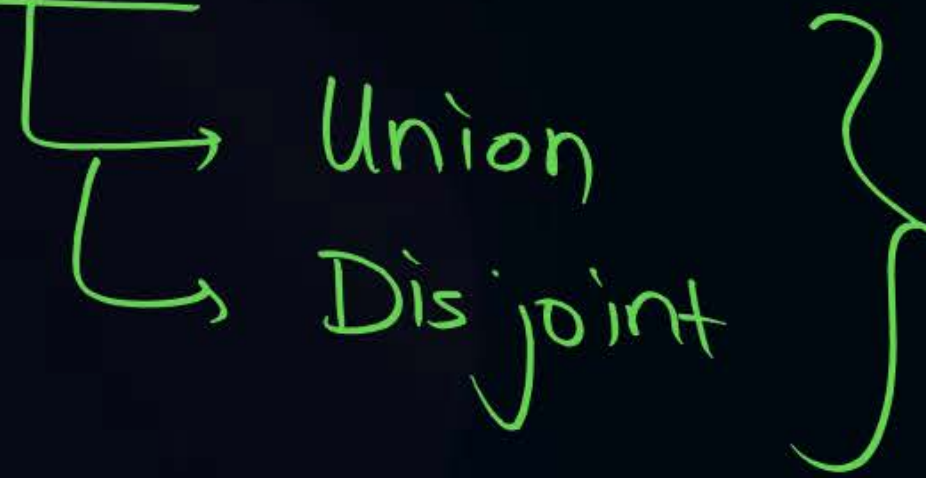


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7. Sets

1. Representations

2. Operations



Kruskal Algo
(must)



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9. Backtracking & Branch- Bound



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Reference Books:

1. Introduction to Algorithm → Cormen
2. Fundamentals of Algorithm → Sahni



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Scope/ Outcomes

GATE, TIFR, ISRO, ^{BARC}~~BASIC~~.....

Placement → Product/service based

Google

Infosys

Microsoft

WIPRO

Accenture

Coding Tests

30th Jan 2019 → Amazon Coding test

3rd Feb 2019 → GATE

5th Feb 2019 → Amazon Interviews ()



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Pre-Requisites

1. Data Structures Fundamental

- [Stacks, Ques, Tree...]

2. Programming Fundamental

- If-~~else~~
- Loops

3. Basic Maths

Series (AP, GP)

Recursion

A) 2nd yr $\rightarrow$ 6%

B) 3rd $\rightarrow$ 33%

C) 4th yr $\rightarrow$ 73%

D) graduation $\rightarrow$ 38%



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Algorithm:-

An Algorithm is a collection of finite number of instruction to solve a given problem.

These instruction are fundamental and should follow a proper sequence.

↓
(atomic)
 $a = b + c$



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Input (i/p) →

Algorithm
(logic)



Output (o/p)

3ex0 [An algorithm may take
or more i/ps]

[Must always produce
at least one output]



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Algo AJSir () $\rightarrow$ 0 inputs

{

print ("Hello students!")

}

Algo AJsir2()

{

return 100 ;

}



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Python:- for i in range (0, n+1)

C++:- For (i = 0, $i \leq n$, i++)

Program → Algorithm implemented using some programming language

Algorithm → Pseudo code - for $i \rightarrow 0 : n$
for $i = 0$ to n

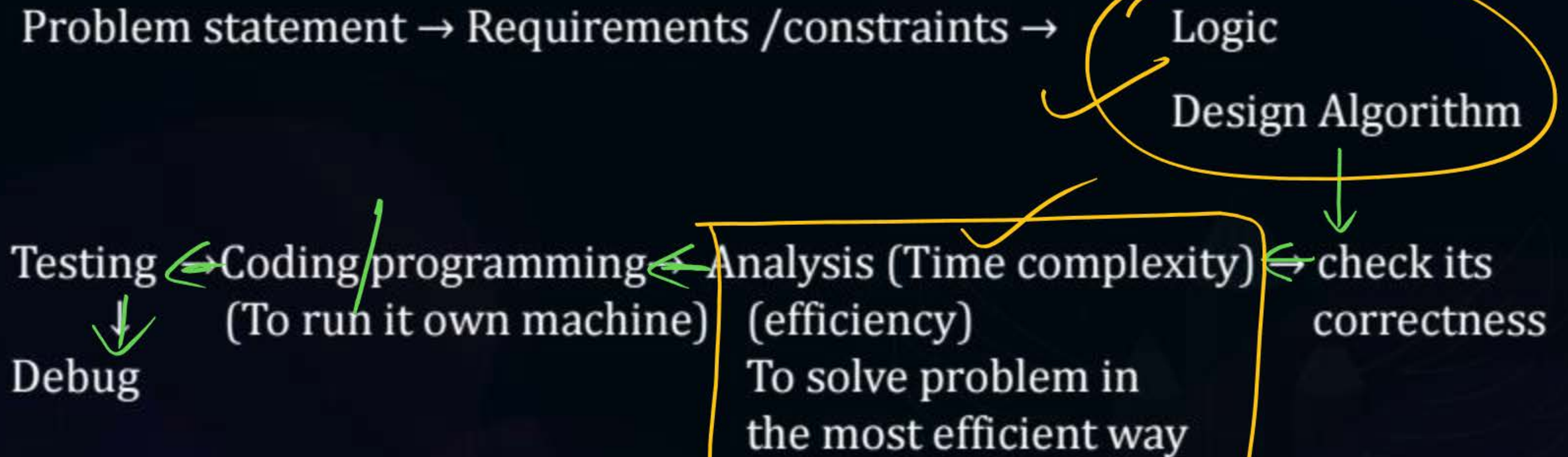
logic

(Set of sequential rules / statement / instructions)



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Algorithm Lifecycle



Analysis of an Algo

- 
- 1) Why?
 - 2) What?
 - 3) How?



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Analysis of Algorithm:-

✓ 1. Why to analysis? → Compare two Algo. ✓

✓ 2. What to analysis? → on the basis of time/space {~~Resudece consume point~~ *reduce consumption*}

3. How to analysis?



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#Q. Why to Analysis ?

To compare and decide which ^{Solution} ~~sudation~~ is the best among different available solution (Algo.)

Problem :- Nagpur → Delhi

Best Solution?

✓ 1. Walk → Soln. (A1)

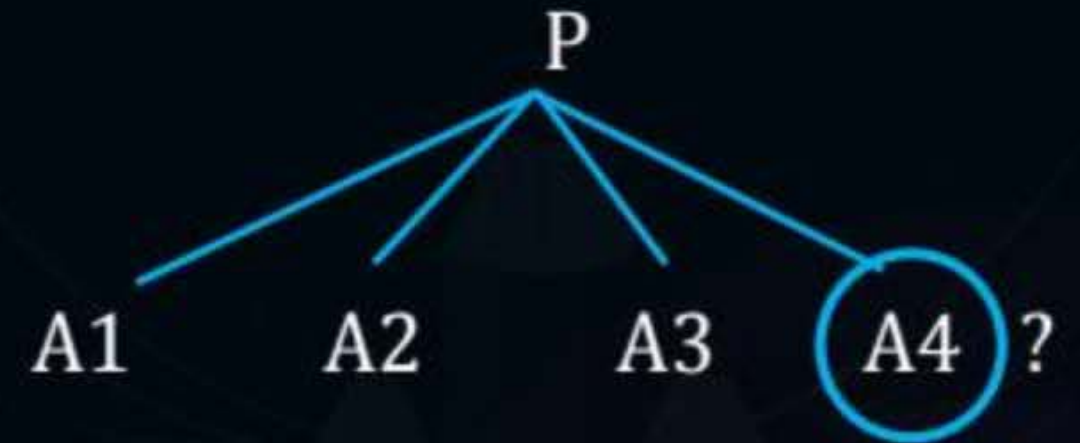
2. Bike / Car → (A2)

3. Train → (A3)

✓ 4. Flight → (A4)

Time

Cost



Time ↓ cost ↑ (Money)



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To Have
Maggie (P)

Self

Cook
(A1)

Chef
(A2)

Zomato
(A3)

Visit Restaurant
(A4)



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#Q. What to Analysis? $\longrightarrow$ metric

Analysis is the consumption of Resources

Resources – Time, space /memory $\longrightarrow$ GATE

Money, Internet Bandwidth, etc.

Number of programmers.

No. of interns

Time $\longrightarrow$ 90%
Space $\longrightarrow$ 10%



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#Q. How to Analysis ?

How to Analysis ?

1) Aposteriori Analysis

Analyzing After
the Implementation of Algo.

2. Apriori Analysis

Analyzing before
the implementation of Algo.

h
Nagpur → Noide Flight

hh: mm: ss ✓

Sam Code/Program

Anandh

Ayushi

50 sec

30 sec

i3



10 sec

30 sec

MCA 10 OS

windows 7

Sam

Sam

Problem Statement



Algo

Anirudh

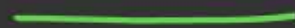
Ayushi



30sec



15sec



Python



C++



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1. Aposteriori Analysis

Advantages

- Given the exact measurement of time units.

High level lang → User Programm friendly

Low level lang → Machine friendly

low faster than High

Disadvantage

- Platform dependent

Software → OS (Linux, windows MAC)

Hardware → (Process /CPU memory (RAM))

Programming language dependent



Summary



- Intro
- Syllabus
- Outcomes
- Why, what, How





THANK - YOU